

Digital self-comb transition boundaries — np.diff

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What this is

Six files that bracket the three genuine on/off transitions of the **digital self-comb** (1.953125 MHz = exactly 8.000 channels, channel-locked, residue 0 mod 8) in the early part of its 07-17/18 era. Each pair is an `ON -> off` file followed by the `off -> ON` file that recovers:

#	ON → off	off → ON
1	<code>corr_20260717_154410Z.h5</code> (15:44:10)	<code>corr_20260717_154619Z.h5</code> (15:46:19)
2	<code>corr_20260717_155704Z.h5</code> (15:57:04)	<code>corr_20260717_155912Z.h5</code> (15:59:12)
3	<code>corr_20260717_161623Z.h5</code> (16:16:23)	<code>corr_20260717_162916Z.h5</code> (16:29:16)

These are the three transitions attributed to the comb itself. The other 17 box-air boundaries in the era are daemon transitions, a daemon outage, or rfswitch/VNA states where the antenna was not connected — i.e. the instrument stopped looking, not the comb stopping. Those are deliberately excluded.

Purpose: visually confirm and characterise what actually changes at these boundaries. This notebook shows the data; it does not settle the comb's physical origin.

Provenance

- Boundary determination: `marjum-2026-07/flags/diag/comb_inventory.jsonl` (best-phase tooth contrast at 8.000 ch, `lock8`), operating point `lock8 >= 30` with phase 0.

- Data access: `eigsep_data.index.MetadataIndex` + `eigsep_data.bundle.load_bundle` — the canonical loaders. The cross correlation is available through `antenna="cross"`, which resolves the `"04"` key per file and fixes conjugation orientation, so there is no reason to drop to raw `h5py` here.
- Units: raw correlator counts throughout. Frequencies in MHz, times UTC.

numpy 2.2.6

Load

`patterns=` restricts the index to just these six files, and `cache=False` keeps the notebook from writing an `.eigsep_index.h5` sidecar into the shared campaign data directory. Both are deliberate: this is a six-file study and it should leave no trace in shared state.

```
indexed 1440 integrations from 6 files
skipped: none
--- box-gnd (auto, input 0) ---
1440 rows x 1024 channels (0.000-249.756 MHz)
antenna 'box-gnd' via keys ['0']
6 files used, 0 skipped
dtype int32

--- box-air (auto, input 4) ---
1440 rows x 1024 channels (0.000-249.756 MHz)
antenna 'box-air' via keys ['4']
6 files used, 0 skipped
dtype int32

--- cross (box-gnd x box-air) ---
1440 rows x 1024 channels (0.000-249.756 MHz)
antenna 'cross' via keys ['04']
6 files used, 0 skipped
dtype complex128
```

Timebase

Every one of these six files has `sync_consistent = True` , so the index's `time_best` is trustworthy here and is used directly for the UTC labels below.

That is worth stating explicitly because it is not generally true of this campaign: `header/times` is corrupt in 642 of 5,120 files (May-2026 dates, negative spans), and the standing rule is to fall back to the filename close time. These six need no such fallback — checked, not assumed.

	role	n_int	sync_ok	utc_start	utc_end	lock8_box_air
file						
corr_20260717_154410Z.h5	ON->off	240	True	15:42:03	15:44:11	0.0
corr_20260717_154619Z.h5	off->ON	240	True	15:44:11	15:46:20	164.6
corr_20260717_155704Z.h5	ON->off	240	True	15:54:56	15:57:04	19.9
corr_20260717_155912Z.h5	off->ON	240	True	15:57:05	15:59:13	136.2
corr_20260717_161623Z.h5	ON->off	240	True	16:14:15	16:16:24	0.0
corr_20260717_162916Z.h5	off->ON	240	True	16:27:08	16:29:17	64.6

all six files: `sync_consistent = True`

int32 accumulator wrap — repaired on the autos only

The correlator's auto accumulators are int32 and **wrap**. Autos are non-negative by construction, so `raw < 0` detects a wrap with no threshold (`eigsep_data.flagging.detectors.overflow_mask`). In this six-file window box-gnd has ~0.22% wrapped samples and box-air ~0.001%.

This matters more here than almost anywhere else: a wrap is a $\pm 2^{32}$ step, and **this notebook differences along time**, so a single unrepaired wrap produces a 4×10^9 -count spike that sets the colour scale and hides everything real.

The cross correlation is deliberately left alone. A cross visibility is genuinely signed in both real and imaginary parts, so `raw < 0` carries no information about wrapping there — applying the auto repair to it would corrupt real data.

```
box-gnd wraps repaired: 3252
box-air wraps repaired: 14
cross left as-is (complex, signed): dtype complex128
1024 channels, 0.000-249.756 MHz
```

Differencing

`np.diff` is taken **within each file**, never across a file join. The six files are not contiguous in time (there are gaps of minutes between them), so differencing a naively concatenated stack would manufacture a bright artificial row at every boundary that has no physical meaning.

Each file's 240 integrations therefore yield 239 difference rows, and a row of `NaN` is inserted between files as a visual separator.

For the cross, the displayed quantity is $\Delta|V|$ — the signed change in visibility amplitude — so that all three panels show the same kind of quantity and can share a colour interpretation. (`|ΔV|` , which also responds to pure phase rotation, is a different question and is not what is plotted here.)

stacked diff shape: (1439, 1024) (6 files x 239 diff rows + 5 separators)
antenna-to-antenna difference rows: 1040 of 1439

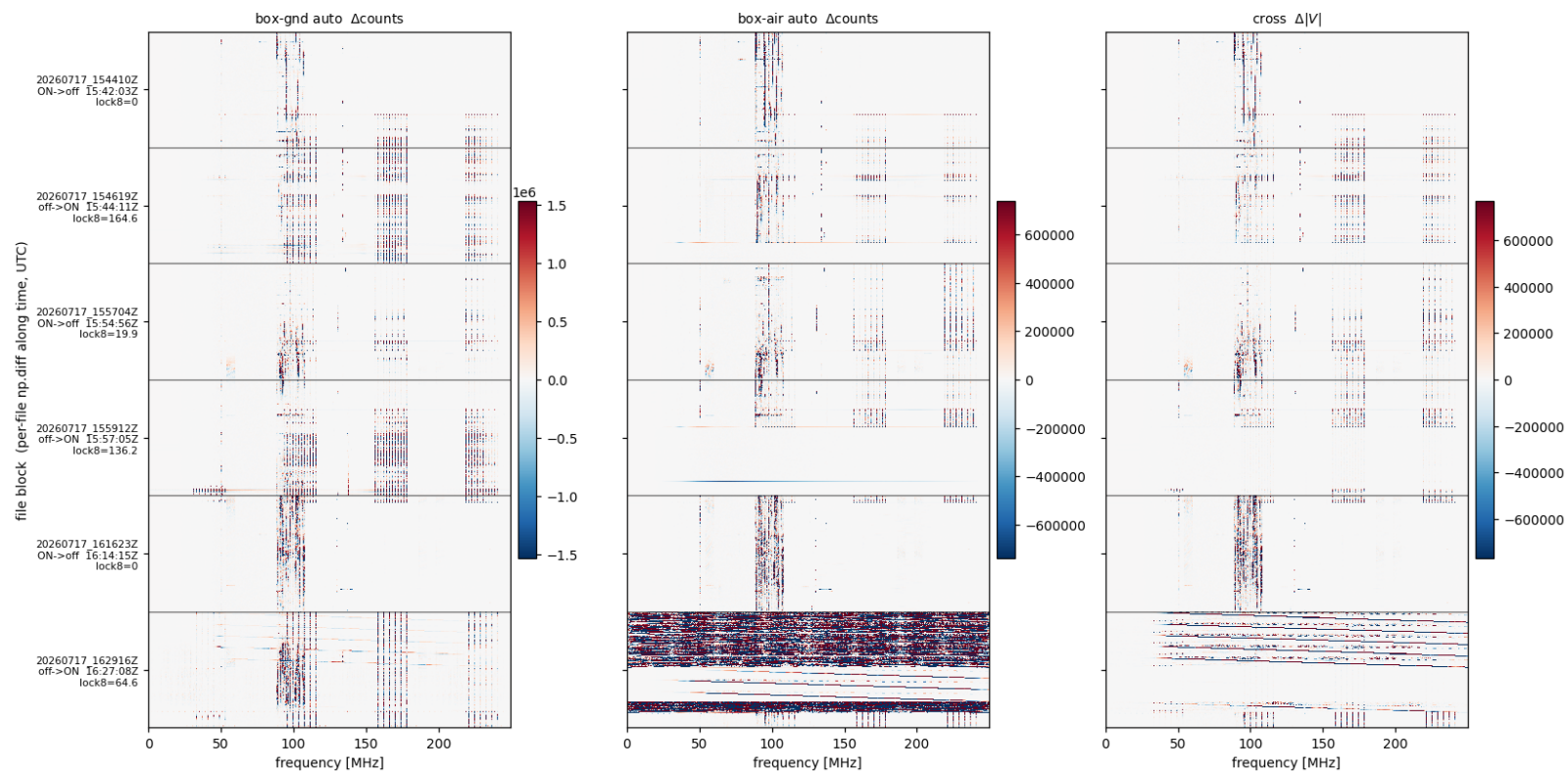
The figure

Three panels, `sharex=True, sharey=True`: box-gnd auto, box-air auto, and the cross amplitude. Frequency runs along x; the concatenated per-file difference rows run down y, with each file's block labelled by name, role and UTC start.

The colour scale is symmetric about zero and set per panel — the three panels differ in absolute scale by orders of magnitude, so a shared colour scale would render two of them blank.

It is set from the **antenna-to-antenna difference rows only**. This is not cosmetic. `corr_20260717_162916Z.h5` runs a VNA sweep, and its switch steps reach $\sim 10^8$ counts — two orders of magnitude above anything on sky. Scaling to all rows lets that one file's calibration saturate the box-air panel and render the other five files blank. Rows outside the antenna state are still drawn, and simply clip.

Digital self-comb transition boundaries — time-difference waterfalls
2026-07-17, three ON/off transition pairs (raw counts)



Which rows are on sky

The same y geometry, showing the receiver state behind every difference row. Green is an antenna-to-antenna difference (a real on-sky step); grey is any row with a load, noise-source or VNA state at either end — those rows are switch steps and carry no sky information.

antenna-to-antenna
difference rows

20260717 154410Z
ON->off 15:42:03Z
lock8=0

20260717 154619Z
off->ON 15:44:11Z
lock8=164.6

20260717 155704Z
ON->off 15:54:56Z
lock8=19.9

20260717 155912Z
off->ON 15:57:05Z
lock8=136.2

20260717 161623Z
ON->off 16:14:15Z
lock8=0

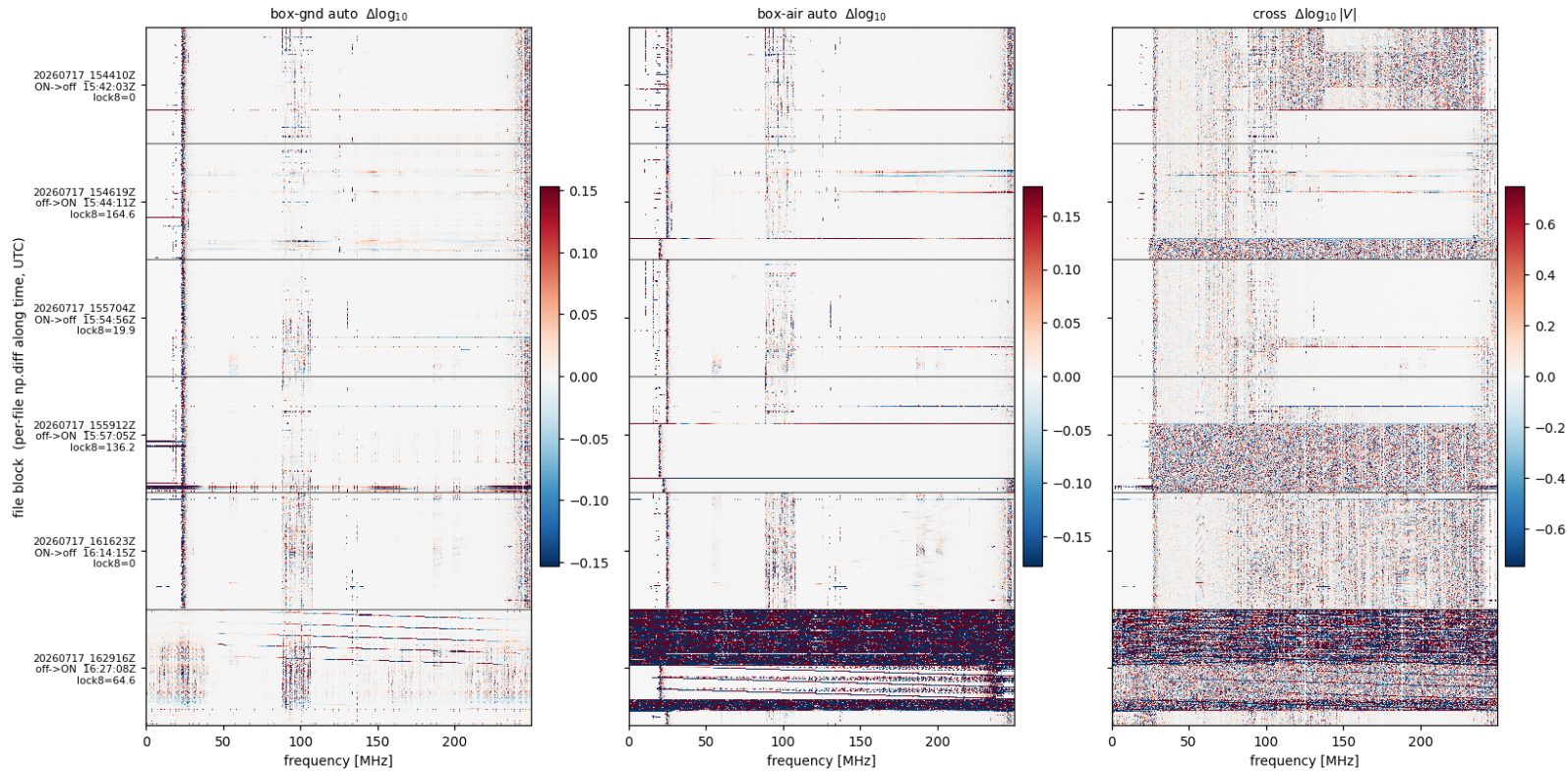
20260717_154410Z: 100.0% on sky
20260717_154619Z: 82.4% on sky
20260717_155704Z: 100.0% on sky
20260717_155912Z: 41.0% on sky
20260717_161623Z: 100.0% on sky
20260717_162916Z: 11.7% on sky

Same difference, log space

The linear panels above are dominated by the brightest channels, which is what a difference of raw counts does. The comb is a narrowband, low-amplitude feature, so the same difference taken in $\log_{10}$ space — i.e. a fractional change per integration — is what actually makes a comb switching on or off legible.

This is the same operation on the same data, not a different selection. The autos use $\log_{10}(\max(\text{counts}, 1))$; the cross uses $\log_{10}(\max(|V|, 1))$.

Same boundaries, difference taken in log space
(fractional change per integration — narrowband features legible)



Read this before interpreting the panels: the rfswitch confound

The receiver switches between the antenna (`RFANT`), an ambient load (`RFAMB`), a noise source (`RFN0N`) and several VNA states on a calibration cadence within these files. Differencing along time responds to every one of those switches, and a switch is a far larger step than the comb.

In this particular six-file selection the cadence is **not** balanced across the two roles:

	state	role	lock8	RFAMB	RFANT	RFNON	UNKNOWN	VNAAMB	VNAL	VNANOFF	VNANON	VNAO	VNARF	VNAS	V
	file														
corr_20260717_154410Z.h5	ON->off		0.0	0	240	0	0	0	0	0	0	0	0	0	
corr_20260717_154619Z.h5	off->ON		164.6	0	198	40	2	0	0	0	0	0	0	0	
corr_20260717_155704Z.h5	ON->off		19.9	0	240	0	0	0	0	0	0	0	0	0	
corr_20260717_155912Z.h5	off->ON		136.2	27	99	110	4	0	0	0	0	0	0	0	
corr_20260717_161623Z.h5	ON->off		0.0	0	240	0	0	0	0	0	0	0	0	0	
corr_20260717_162916Z.h5	off->ON		64.6	0	29	0	12	22	22	22	22	22	22	22	

```

pure-RFANT (no cal switching in the file):
corr_20260717_154410Z.h5  role=ON->off  pure_RFANT=True
corr_20260717_154619Z.h5  role=off->ON  pure_RFANT=False
corr_20260717_155704Z.h5  role=ON->off  pure_RFANT=True
corr_20260717_155912Z.h5  role=off->ON  pure_RFANT=False
corr_20260717_161623Z.h5  role=ON->off  pure_RFANT=True
corr_20260717_162916Z.h5  role=off->ON  pure_RFANT=False

```

All three ON -> off files are 100% RFANT . All three off -> ON files contain calibration or VNA states. In this sample the split is perfect, 3/3 and 3/3.

That is a confound, and it has to be stated plainly: within these six files alone, "comb detected" and "calibration cadence running" are not separable, so nothing in the figures above can by itself distinguish the comb switching on from the calibration hardware switching on.

It is, however, a property of this small selection and **not** of the era. Across the full 07-17 15:37 → 07-18 03:00 era on box-air, at the same lock8 >= 30 operating point, the association runs the other way:

	pure RFANT	cal/VNA active
comb ON	176	29
comb off	49	56

So era-wide the comb is predominantly detected in clean antenna files, and the six boundary files here are an unrepresentative stretch. The honest reading is that these six make a good visual case study of what a boundary looks like, and a **bad** basis for attribution.

One further caveat specific to corr_20260717_162916Z.h5 : only 29 of its 240 integrations are RFANT , so its lock8 = 64.6 detection rests on 29 antenna integrations, not 240.

Decision requested

1. Does the characterisation above match what you expect a genuine comb on/off boundary to look like, or do the panels instead read as calibration switching to you?
2. If the rfswitch confound is the dominant concern, the natural next step is to re-pick boundary files **matched on cal state** (both sides of a transition pure- RFANT) and repeat this figure. Era-wide there are 49 pure- RFANT comb-off files to draw from. That is a new selection and a new milestone, so I have not done it here.
3. `lock8 >= 30` is the operating point used to call these transitions; it is defensible (a stable plateau, and 216/220 agreement with beam-analyst's independent detector) but it is a choice, and transition counts move with it.

Not claimed here: anything about the comb's physical origin. This notebook shows morphology at six boundaries; it runs no pointing test, no null control, and no far-field discriminator.